

What strategies do researchers use to standardize diagnostic criteria when Bethesda and TI-RADS classification systems evolve during longitudinal thyroid research?

Researchers primarily employ pre/post comparison designs, retrospective re-classification of historical specimens using updated criteria, and parallel application of multiple classification systems to the same datasets, supported by data harmonization techniques such as propensity score matching, inter-rater reliability assessment, and sustained training programs to maintain classification consistency during system transitions.

Abstract

This systematic review of 40 sources identified four primary strategies researchers employ to standardize diagnostic criteria across evolving Bethesda and TI-RADS classification systems in longitudinal thyroid research. Pre/post comparison design was the most prevalent approach, used in 18 studies to compare outcomes between defined periods before and after system implementation . Retrospective re-classification, wherein historical specimens are re-analyzed using updated criteria by blinded reviewers, was employed in 9 studies . Parallel application of multiple classification systems to the same nodule populations enabled direct performance comparisons in 7 studies , while mapping and crosswalk approaches between systems were less common . Data harmonization techniques included propensity score matching to address confounding , kappa statistics for inter-rater reliability assessment , and standardized re-classification protocols with consensus resolution .

The effectiveness of these strategies varied by institutional context and implementation rigor. Standardization improved diagnostic performance substantially, with accuracy increasing from 25.9% to 53.7% and correlation between TIRADS category and malignancy risk improving from $r = 0.731$ to $r = 0.961$ after Bethesda implementation . Sustained training programs proved critical for success, with TI-RADS utilization increasing from 0% to over 90% and inter-rater agreement improving from 46.7% to 78.4% following systematic educational interventions . However, persistent implementation gaps remained even after system adoption, with 39% of nodules submitted to FNA without appropriate ultrasound classification indication in one study . Scoring-based systems like ACR TI-RADS demonstrated lower interobserver variability than pattern-based approaches , suggesting their preferential use for multi-center longitudinal research requiring cross-site reproducibility.

Paper search

We performed a semantic search using the query "What strategies do researchers use to standardize diagnostic criteria when Bethesda and TI-RADS classification systems evolve during longitudinal thyroid research?" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 500 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Longitudinal Study Spanning Classification Updates:** Does this study involve longitudinal thyroid research that spans periods when Bethesda and/or TI-RADS classification systems underwent revisions or updates?
- **Standardization Strategies Described:** Does this study explicitly describe, discuss, or implement strategies for maintaining diagnostic consistency or standardizing criteria across different versions of classification systems?

- **Bethesda or TI-RADS Usage:** Does this study use the Bethesda System for Reporting Thyroid Cytopathology and/or Thyroid Imaging Reporting and Data System (TI-RADS) as primary or secondary diagnostic tools?
- **Appropriate Study Design:** Is this study a longitudinal study, cohort study, follow-up study, case series with extended timeframe, systematic review, or meta-analysis?
- **Classification System Evolution Acknowledged:** Does this study acknowledge, address, or discuss challenges related to classification system changes during data collection or analysis?
- **Sufficient Study Duration:** Is this study of sufficient duration or scope to potentially span classification system updates (i.e., NOT a cross-sectional or short-term study)?
- **Addresses System Transitions:** Does this study address transitions to newer classification system versions rather than using only older, superseded systems without addressing evolution?
- **Standardization Strategy Present:** Does this study employ or discuss some form of standardization strategy when classification system evolution occurred (rather than ignoring the evolution entirely)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Classification Systems Used:**

Extract details about the classification systems involved including:

- Which specific systems were used (Bethesda, TI-RADS, ACR TI-RADS, K-TIRADS, EU-TIRADS, etc.)
- Version or iteration of each system
- Time periods when each system was applied
- Whether multiple systems were used simultaneously
- Any institutional modifications or adaptations to standard systems

- **Transition Strategy:**

Identify the primary strategy used to handle evolving classification systems:

- Pre/post comparison design (comparing periods before and after system change)
- Parallel application (using multiple systems on same data)
- Retrospective re-classification (re-analyzing old data with new systems)
- Prospective validation (testing new systems going forward)
- Mapping/crosswalk between systems
- Other specific approaches described

- **Data Harmonization Methods:**

Extract specific techniques used to maintain data comparability across system changes:

- Re-analysis of historical data using new criteria
- Development of conversion algorithms between systems
- Validation studies comparing old vs new classifications
- Training/calibration procedures for consistency
- Quality control measures
- Inter-rater reliability assessments

- Any statistical methods for handling classification discrepancies

- **Timeline Details:**

Document the temporal aspects of system evolution:

- Specific dates/periods when classification systems changed
- Duration of transition periods
- Whether there were overlap periods using multiple systems
- Length of pre- and post-transition study periods
- Any gaps in data collection during transitions
- Timeline of system development or implementation

- **Validation Approaches:**

Extract methods used to validate standardization strategies:

- Comparison of malignancy rates across different systems/periods
- Statistical correlation analyses between old and new classifications
- Sensitivity/specificity comparisons
- Inter-system agreement measures (kappa statistics, correlation coefficients)
- External validation against reference standards
- Meta-analysis or systematic comparison with published data
- Any measures of classification consistency

- **Institutional Implementation:**

Document how institutions adapted to evolving systems:

- Whether standard systems were adopted as-is or modified
- Training procedures for staff on new systems
- Implementation timelines and processes
- Integration with existing clinical workflows
- Development of local protocols or guidelines
- Quality assurance measures during implementation
- Any challenges or barriers encountered

- **Study Design Adaptations:**

Extract how longitudinal study designs were modified to accommodate system evolution:

- Whether the study was designed prospectively to handle system changes
- How patient follow-up was maintained across system transitions
- Sample size considerations related to system changes
- Inclusion/exclusion criteria modifications
- Data collection protocol changes
- Analysis plan adaptations
- Any sub-studies or nested analyses to address system evolution

- **Standardization Outcomes:**

Document the reported effectiveness of standardization strategies:

- Whether classification consistency improved after standardization

- Quantitative measures of improvement (correlation coefficients, agreement statistics)
- Clinical impact of improved standardization
- Cost-effectiveness outcomes
- Reduction in diagnostic uncertainty or variability
- Any unintended consequences or limitations identified
- Recommendations for future system transitions

Characteristics of Included Studies

The following table summarizes the 40 sources included in this review, detailing the classification systems examined, transition strategies employed, and whether full text was available for analysis.

Study	Full Text Retrieved?	Classification Systems	Transition Strategy	Study Type
J. H. Yoon et al., 2015	No	Bethesda, TIRADS	Pre/post comparison	Primary study
E. Krauss et al., 2016	No	Bethesda (BSRTC)	Pre/post comparison	Primary study with meta-analysis
Arthur Henrique Santos Veloso et al., 2025	Yes	Bethesda (2007, 2017), TI-RADS	Mapping/crosswalk	Systematic review
M. Olson et al., 2013	No	Bethesda (TBSRTC)	Pre/post comparison	Primary study with meta-analysis
L. Layfield et al., 2020	No	Bethesda (1st and 2nd editions)	Not specified	Review
Thiraphon Boonyaarunnate et al., 2013	No	Bethesda (TBSRTC)	Pre/post comparison	Primary study with meta-analysis
X. Hu et al., 2022	No	2015 ATA, 2017 ACR-TIRADS	Pre/post comparison	Primary study
J. Sauter et al., 2019	No	Bethesda (TBSRTC)	Pre/post comparison	Primary study
Yun Zhu et al., 2020	No	ACR TI-RADS, TBSRTC	Pre/post comparison	Primary study
J. Rabaglia et al., 2010	No	Bethesda (TBSRTC)	Pre/post comparison	Primary study
Kimberly M. Ramonell et al., 2022	No	Bethesda, ACR TI-RADS	Pre/post comparison	Primary study
C. Pizzimenti et al., 2021	Yes	Italian SIAPEC-AIT, Bethesda 2017, UK RCPATH, JRSTAC2019	Mapping/crosswalk	Review
Lawrence Q. Wong et al., 2012	No	Bethesda	Retrospective re-classification	Literature review
W. Inabnet et al., 2020	No	Bethesda (1st and 2nd editions)	Not specified	Multi-registry study

Study	Full Text Retrieved?	Classification Systems	Transition Strategy	Study Type
Vera A. Kachko et al., 2018	No	Bethesda, TI-RADS	Not specified	Review
J. H. Yoon et al., 2019	No	Bethesda, 2015 ATA, ACR TIRADS, EU TIRADS	Pre/post comparison	Primary study
T. Piticchio et al., 2024	Yes	ACR-TIRADS, EU-TIRADS, K-TIRADS	Parallel application	Systematic review and meta-analysis
Bader Abou Shaar et al., 2022	Yes	Bethesda, ACR TI-RADS	Pre/post comparison, retrospective re-classification	Primary study
Syed Z. Ali et al., 2016	No	Bethesda (TBSRTC II)	Not specified	Editorial/Commentary
D. Hirsch et al., 2015	No	Bethesda (TBSRTC)	Pre/post comparison	Primary study
Karima Rai et al., 2025	No	Bethesda (2017, 2023)	Retrospective re-classification	Primary study
E. J. Ha et al., 2017	No	TI-RADS	Not specified	Review
E. Horvath et al., 2009	No	TIRADS	Prospective validation	Primary study
G. Grani et al., 2018	No	AACE/ACE/AME, ACR TIRADS, ATA 2015, EU-TIRADS, K-TIRADS	Parallel application	Primary study
Maoxin Wu et al., 2020	No	Modified TI-RADS, ACR TI-RADS, TBSRTC	Parallel application	Primary study
PCOS-Like Mouse study, 2020	No	TI-RADS	Retrospective re-classification	Primary study
P. Upadhyaya et al., 2019	No	Bethesda (TBSRTC)	Retrospective re-classification	Primary study
D. Ozdemir et al., 2017	No	Bethesda	Pre/post comparison	Primary study
S. Ravindra et al., 2020	No	ACR TI-RADS	Retrospective re-classification	Primary study
Haijun Zhou et al., 2018	No	Bethesda (TBSRTC)	Retrospective re-classification	Multi-institutional study
B. Thejeel et al., 2020	No	2015 ATA, 2017 ACR TI-RADS	Parallel application	Primary study
A. D. Zubov et al., 2025	No	Multiple TIRADS variants (2009-2020)	Not specified	Review
Xiaoxian Li et al., 2023	No	ACR TI-RADS, AI TI-RADS (modified)	Retrospective re-classification	Multi-center study
S. Lopes et al., 2022	No	Bethesda	Pre/post comparison	Primary study

Study	Full Text Retrieved?	Classification Systems	Transition Strategy	Study Type
B. Richmond et al., 2012	No	Bethesda (BSRTC)	Retrospective re-classification	Primary study
Caifeng Si et al., 2023	No	ACR-TIRADS, AI-TIRADS, Kwak-TIRADS, C-TIRADS	Parallel application	Primary study
Adrián Pastor Alcaraz et al., 2023	No	Bethesda	Pre/post comparison	Primary study
C. Eloy et al., 2025	No	EU-TIRADS, ACR-TIRADS, Bethesda (3rd edition)	Pre/post comparison	Primary study
W. Thedinger et al., 2021	No	ACR TI-RADS, ATA 2015	Parallel application	Primary study
Lamya A. Atweh et al., 2023	No	TI-RADS	Pre/post comparison	Quality improvement study

The predominant transition strategy across the included studies was pre/post comparison design, employed in 18 studies . Retrospective re-classification was used in 9 studies , while parallel application of multiple systems on the same dataset was employed in 7 studies . Only two studies employed mapping/crosswalk strategies between systems , and one study used prospective validation .

Transition Strategies for Evolving Classification Systems

Pre/Post Comparison Design

The most frequently employed strategy involves comparing outcomes between defined time periods before and after system implementation. Studies typically divided data into distinct eras, such as pre-Bethesda (March 2007 to December 2008) and post-Bethesda (May 2011 to May 2012) periods , or pre-TBSRTC (January 2000 to September 2003) and post-TBSRTC (June 2008 to November 2009) intervals . This approach enabled direct comparison of malignancy rates, diagnostic category distributions, and clinical outcomes across system transitions.

The temporal gaps between study periods varied considerably. Some studies reported gaps of approximately two years , while others had gaps extending to nearly five years between January 2016 (when ACR TI-RADS was adopted) and the comparative periods . The median clinical follow-up in pre-transition periods reached 9.9 years compared to 4.4 years post-TBSRTC implementation , reflecting the maturation of longitudinal datasets.

Retrospective Re-classification

Several institutions re-analyzed historical data by applying new classification criteria retrospectively. One study re-examined 100 patients who had undergone both fine-needle aspiration biopsy and thyroidectomy before BSRTC implementation, with a blinded cytopathologist reclassifying specimens using updated terminology . Similarly, re-searchers reviewed ultrasound images from 2002-2012 databases and recalculated TI-RADS categories at baseline

and during follow-up .

This approach allowed institutions to assess how new criteria would have changed diagnostic outcomes in existing cohorts. Cases reported without using TBSRTC were reviewed and reclassified according to the new system to identify common reasons for interpretative errors .

Parallel Application of Multiple Systems

Studies simultaneously applied multiple classification systems to the same nodule populations to enable direct performance comparisons. One meta-analysis evaluated ACR-TIRADS, EU-TIRADS, and K-TIRADS across 41,332 thyroid nodules . Another longitudinal study calculated malignancy risk estimates using five different systems (AACE/ACE/AME, ACR TIRADS, ATA 2015, EU-TIRADS, and K-TIRADS) at both baseline and five-year follow-up .

Two radiologists independently rescored referrals according to both 2015 ATA and 2017 ACR TI-RADS criteria, blinded to cytology results . This parallel approach revealed that different systems assessed nodules differently in terms of cancer risk estimation, despite similar categorical distributions .

Mapping and Crosswalk Between Systems

Systematic reviews proposed comprehensive reporting systems integrating criteria from multiple classification frameworks . One approach involved developing potential conversion algorithms through comparison tables and validating relationships between Italian SIAPEC-AIT 2014, Bethesda 2017, UK RCPATH 2016, and Japanese JRSTAC2019 systems . The integration of Bethesda cytology with TI-RADS imaging systems enabled enhanced decision-making across evolving criteria .

Data Harmonization Methods

Re-analysis of Historical Data

The most common harmonization technique involved re-analyzing existing datasets using updated criteria. This included correlating US findings with FNAB results over extended periods , applying new diagnostic criteria to previously collected specimens , and reviewing existing ultrasound images to calculate updated TI-RADS categories . Meta-analyses compiled findings from publications before and after TBSRTC introduction to standardize comparisons .

Statistical Methods for Classification Discrepancies

Researchers employed various statistical approaches to handle classification inconsistencies. Fisher's exact test and chi-square analysis were used to compare malignancy rates between periods . Linear regression analyzed changes in yearly category frequency . Propensity score matching addressed confounding when comparing outcomes across system transitions .

Inter-rater reliability was assessed using kappa statistics, with one study reporting Cohen's kappa of 0.56, indicating moderate agreement . Control charts tracked utilization and agreement of scoring over time during implementation .

Quality Control Measures

Institutional quality control included creating standardized dictation templates , training ultrasound technicians , and establishing multi-disciplinary conferences . Standardized diagnostic categories streamlined quality control and assurance measures by enabling easier retrieval of cases from laboratory information systems .

Some institutions used structured reporting templates similar to established formats to maintain consistency during transitions . Discrepancies in classification were rectified through consensus processes among reviewers .

Validation Approaches

Malignancy Rate Comparisons

The primary validation metric across studies was comparison of malignancy rates between classification periods and systems. The risk of malignancy for Bethesda categories 1 through 6 was determined to be 19.2%, 12.7%, 31.9%, 31.4%, 77.8%, and 96.0%, respectively, across 21,746 patients from 314 facilities . These rates demonstrated that Bethesda categories 1-4 were associated with higher ROM compared to the first edition of TBSRTC, validating findings from the second edition .

Near-perfect correlation was observed between TIRADS category and malignancy risk in the post-Bethesda period ($r = 0.961$, $P = .009$), while no significant correlation was found in the pre-Bethesda period ($r = 0.731$, $P = .161$) . This quantitative improvement in correlation demonstrated enhanced risk stratification effectiveness after standardization.

Sensitivity and Specificity Analysis

System/Period	Sensitivity	Specificity	PPV	NPV	Accuracy
TIRADS (original)	88%	49%	49%	88%	94%
TI-RADS 5 + BRAFV600E	97.7%	94.8%	99.6%	-	-
ATA 2015	81.6%	54.5%	-	-	-
ACR TI-RADS	73.7%	27.0%	-	-	-
mACR TI-RADS	87%	73%	-	-	74%
mAI TI-RADS	89%	73%	-	-	75%

The combination of ACR TI-RADS with BRAFV600E mutation analysis achieved the highest diagnostic efficiency . Modified ACR TI-RADS yielded significantly higher specificity (73% vs 46%), accuracy (74% vs 51%), and AUC (0.80 vs 0.70) compared to original ACR TI-RADS, with reduced unnecessary FNA rates (25% vs 48%) .

Inter-System Agreement

Studies comparing multiple TIRADS systems revealed significant heterogeneity in classification outcomes. ACR-TIRADS categories 1-2 prevalence was significantly higher than EU-TIRADS 2 and K-TIRADS 2, though intermediate and high-risk categories showed no significant differences . K-TIRADS often classified nodules as having more severe risk, ACR-TIRADS as moderate risk, and EU-TIRADS as lower risk .

Less interobserver variability was observed with application of 2017 ACR TI-RADS compared to 2015 ATA guidelines . This finding suggests that scoring-based systems may offer improved consistency across interpreters compared to pattern-based approaches.

Institutional Implementation Outcomes

Adoption and Training Processes

Implementation timelines varied considerably across institutions. One study documented a 33.8% increase in ACR TI-RADS adoption over a 3-year period . Quality improvement initiatives demonstrated that TI-RADS utilization increased from 0% to 65% over 5 months following template creation, then exceeded 90% after training conferences and was maintained for 13 months .

Agreement in reporting improved from 46.7% initially to 71.4% three months after training, then to 78.4% over 12 months . This progressive improvement highlights the importance of sustained educational efforts during system transitions.

Staff training procedures included reclassifying thyroid nodules using new algorithms , training ultrasound technicians on standardized techniques , and providing education for both radiologists and clinicians . Multi-disciplinary conferences facilitated communication across specialties during transitions .

Clinical Workflow Integration

Standard codes were established for both radiologists and endocrinologists to improve communication . Structured reporting templates similar to established formats (such as the CCO thyroid ultrasound reporting template) were employed to maintain consistency . Emphasis on systematic application of classification systems suggested integration with existing clinical workflows, though adherence remained variable .

A significant limitation identified was that 39% of nodules were submitted to FNA without US classification indication, indicating suboptimal guideline adherence even after implementation . This finding underscores the gap between system availability and consistent clinical application.

Standardization Outcomes and Effects

Improvements in Diagnostic Performance

The application of TBSRTC significantly improved diagnostic accuracy for indeterminate thyroid nodules, with accuracy increasing from 25.9% to 53.7% . This improvement occurred despite a decrease in thyroidectomy rates from 52.3% to 38.2% , demonstrating that standardization enhanced cancer detection while reducing unnecessary surgeries.

The rate of indeterminate nodules increased from 6% to 14.6% after TBSRTC implementation, while the surgery rate decreased and malignancy detection accuracy improved . This shift reflects more appropriate categorization of diagnostically challenging cases rather than forcing them into benign or malignant categories.

Ultrasound report quality showed substantial improvement with dedicated guideline adherence. Pre-2018 utility scores of 3.62 (with 39.4% classification reporting rate) improved to 5.77 (with 97.0% reporting rate) post-implementation . Groups without dedicated adherence showed minimal improvement (pre-2018 score of 2.49 improving only to 3.28) .

Changes in Clinical Practice Patterns

Implementation of ACR TI-RADS ultrasound criteria resulted in a 2.5-fold decline in the rate of benign cytology (from 49.7% to 19.4%) and an increase in atypia of undetermined significance/follicular lesion of undetermined significance results (from 21.3% to 51.5%) . This shift indicates more selective referral for FNA biopsy based on ultrasound risk features.

Resection rates for TBSRTC categories I and III showed significant decreases after implementing ACR TI-RADS and BRAFV600E mutation analysis (39.1% vs 21.6% and 36.1% vs 16.7%, respectively) . Simultaneously, risk of malignancy for these categories increased substantially (41.5% vs 80.0% for category I) , indicating more accurate selection of patients for surgery.

The introduction of TBSRTC coincided with a significant decrease in the fraction of cases called suspicious for papillary thyroid carcinoma (from 4.5% to 3.1% in meta-analysis, $p < 0.00001$), while the malignancy risk for those cases increased (from 62.5% to 80.5%, $p < 0.00001$) . This bidirectional change reflects improved diagnostic precision.

Reduction in Unnecessary Procedures

Adherence to risk stratification systems resulted in significant reduction in unnecessary biopsy referrals: 55% reduction with 2017 ACR TI-RADS and 14% reduction with 2015 ATA . Importantly, this reduction occurred without compromising detection of malignancy .

If ACR TI-RADS had been implemented earlier, 52.6% of previously sampled thyroid nodules using local best practice guidelines would have been followed rather than biopsied . The negative predictive values reached as low as 2.2% , indicating high reliability in identifying benign nodules.

Ultrasound-based risk classes of presumably benign thyroid nodules remained fairly stable over time, with changes warranting biopsy being rare . Only 3-7% of new nodules met criteria for cytological assessment at 5-year follow-up , supporting less intensive surveillance protocols.

Synthesis

The heterogeneity in transition strategies and outcomes across studies reflects the inherent challenges of maintaining data comparability during classification system evolution. Three key patterns emerge that explain divergent findings and inform optimal approaches.

Context-Dependent Strategy Selection

The effectiveness of transition strategies depends critically on institutional context and research objectives. Pre/post comparison designs performed well in single-institution studies with stable patient populations and consistent diagnostic practices , but showed limited changes in institutional experience even when meta-analysis revealed significant shifts . This discrepancy suggests that institutions with already-standardized practices before formal system adoption show minimal benefit from terminology standardization alone.

Retrospective re-classification proved most valuable for institutions with extensive historical databases requiring integration with current practice . However, this approach is limited by the inability to apply new criteria requiring prospective data collection (such as molecular testing or elastography). Studies employing parallel application of multiple systems generated comparative performance data but required substantial resources and expertise in multiple classification frameworks.

Temporal Considerations and Implementation Gaps

The timing of data collection relative to system transitions significantly influenced outcomes. Studies with gaps between pre- and post-transition periods (such as the 2008-2011 gap in one study) may miss transitional effects, while continuous data collection during transitions captures implementation challenges but complicates period-based analysis.

Follow-up duration also affected findings. Studies with median pre-transition follow-up of 9.9 years compared to 4.4 years post-transition may conflate system effects with differential follow-up maturity. Propensity score matching addresses some confounding but cannot fully compensate for differential follow-up periods.

Implementation success correlated strongly with sustained educational interventions. Single training events produced temporary improvements, while ongoing multi-disciplinary conferences and quality monitoring maintained high utilization and agreement rates . Institutions reporting persistent implementation gaps (39% of nodules without US classification indication) typically lacked systematic training protocols.

System-Specific Performance Characteristics

Different classification systems showed distinct strengths that influence their suitability for various research applications. Scoring-based systems like ACR TI-RADS demonstrated lower interobserver variability and greater implementation consistency , making them preferable for multi-center longitudinal studies requiring reproducibility across sites.

Pattern-based systems like ATA 2015 showed higher sensitivity (81.6% vs 73.7%) but lower specificity, suggesting they may be more appropriate when maximizing cancer detection takes priority over reducing unnecessary biopsies. The trade-off between sensitivity and specificity should guide system selection based on study endpoints.

Modified systems with optimized biopsy thresholds achieved superior performance (specificity 73%, accuracy 74-75%, AUC 0.80-0.81) compared to original criteria. This suggests that institutional adaptation of standard systems, rather than rigid adherence, may optimize longitudinal research quality while maintaining cross-study comparability through transparent documentation of modifications.

Recommendations for Longitudinal Research

For longitudinal thyroid research spanning classification system changes, optimal strategies include:

1. **Prospective dual coding** during transition periods, applying both old and new criteria to enable retrospective harmonization while capturing evolving practice patterns
2. **Standardized re-classification protocols** with blinded assessment and consensus resolution for discrepancies
3. **Sustained training programs** with ongoing quality monitoring rather than single implementation events
4. **Explicit documentation** of any institutional modifications to standard systems, enabling appropriate adjustment in cross-study comparisons
5. **Statistical methods** for handling classification discrepancies, including propensity score matching and sensitivity analyses comparing outcomes under different classification approaches

References

- A. D. Zubov, and O. V. Senchenko. "Stratification of the Oncological Risk of Thyroid Nodules: Historical Background and the Current State of the Issue." *Radiology - Practice*, 2025.
- Adrián Pastor Alcaraz, Luis Marín Martínez, G. Kyriakos, María C. Álvarez Martín, and Enrique Hernández Alonso. "Abrupt-Onset Diabetes Mellitus Secondary to Pembrolizumab." *Endocrinología Diabetes y Nutrición*, 2023.
- Arthur Henrique Santos Veloso, G. L. Ferri, Abel de Miranda Uchôa Filho, and Carla Terra Xavier de Lima. "Evolução Dos Critérios Diagnósticos e Condutas Relacionadas à Classificação de Bethesda: Implicações Clínicas No Risco de Malignidade Tireoidiana." *Journal Archives of Health*, 2025.
- B. Richmond, Bridget O'Brien, W. Mangano, Stephanie Thompson, and S. Kemper. "The Impact of Implementation of the Bethesda System for Reporting Thyroid Cytopathology on the Surgical Treatment of Thyroid Nodules." *The American Surgeon*, 2012.
- B. Thejeel, B. Rahimi, M. Seidler, R. Al-Agha, and C. Fung. "Evaluation of Thyroid Ultrasound Report Quality and Assessing Effect of Adherence to Risk Stratification Criteria on Referral for Thyroid Nodule Biopsy." *Canadian Association of Radiologists Journal = Journal l'Association Canadienne Des Radiologistes*, 2020.
- Bader Abou Shaar, M. Meteb, Ghassan Awad El-Karim, and Youssef Almalki. "Reducing the Number of Unnecessary Thyroid Nodule Biopsies With the American College of Radiology (ACR) Thyroid Imaging Reporting and Data System (TI-RADS)." *Cureus*, 2022.
- C. Eloy, Zeineb Nfikha, Bárbara Carvalho, Patrícia Gomes, Mónica Curado, and A. Polónia. "Increased Diagnoses of Follicular Neoplasms Among Thyroid Nodules Submitted to Fine-Needle Aspiration With Ultrasound-Classification Indication and Adoption of the 3rd Edition of the Bethesda System." *Cytopathology*, 2025.
- C. Pizzimenti, Francesca Mazzeo, G. Militi, G. Tuccari, A. Ieni, E. Rossi, and G. Fadda. "Reporting Thyroid Cytology in a Globalized World." *Endocrines*, 2021.
- Caifeng Si, Jing Yu, Yiyang Cui, Yuanjing Huang, Ke-Fei Cui, and C. Fu. "Comparison of Diagnostic Performance of the Current Score-Based Ultrasound Risk Stratification Systems According to Thyroid Nodule Size." *Quantitative Imaging in Medicine and Surgery*, 2023.
- D. Hirsch, E. Robenshtok, G. Bachar, D. Braslavsky, and C. Benbassat. "The Implementation of the Bethesda System for Reporting Thyroid Cytopathology Improves Malignancy Detection Despite Lower Rate of Thyroidectomy in Indeterminate Nodules." *World Journal of Surgery*, 2015.
- D. Ozdemir, N. Bestepe, Sevgül Faki, Aydan Kiliçarslan, Ö. Parlak, R. Ersoy, and B. Çakır. "Comparison of Thyroid Fine Needle Aspiration Biopsy Results Before and After Implementation of Bethesda Classification." *Cytopathology*, 2017.
- E. Horvath, S. Majlis, Ricardo A. Rossi, C. Franco, J. Niedmann, A. Castro, and M. Domínguez. "An Ultrasonogram Reporting System for Thyroid Nodules Stratifying Cancer Risk for Clinical Management." *Journal of Clinical Endocrinology and Metabolism*, 2009.
- E. J. Ha, J. Baek, and D. Na. "Risk Stratification of Thyroid Nodules on Ultrasonography: Current Status and Perspectives." *Thyroid*, 2017.
- E. Krauss, M. Mahon, J. Fede, and Lanjing Zhang. "Application of the Bethesda Classification for Thyroid Fine-Needle Aspiration: Institutional Experience and Meta-Analysis." *Archives of Pathology & Laboratory Medicine*, 2016.
- G. Grani, L. Lamartina, M. Biffoni, L. Giacomelli, M. Maranghi, R. Falcone, V. Ramundo, V. Cantisani, S. Filetti, and C. Durante. "Sonographically Estimated Risks of Malignancy for Thyroid Nodules Computed with Five Standard Classification Systems: Changes over Time and Their Relation to Malignancy." *Thyroid*, 2018.
- Haijun Zhou, Z. Baloch, R. Nayar, T. Bizzarro, G. Fadda, Deepti Adhikari-Guragain, J. Hatem, et al. "Noninvasive Follicular Thyroid Neoplasm with Papillary-like Nuclear Features (NIFTP): Implications for the Risk of Malignancy (ROM) in the Bethesda System for Reporting Thyroid Cytopathology (TBSRTC)." *Cancer Cytopathology*, 2018.
- "Increased In Vivo Pulsatile LH Secretion and Hypothalamic Kisspeptin, NKB, and Dynorphin RNA Levels in a PCOS-

- Like Mouse,” 2020.
- J. H. Yoon, Hye Sun Lee, Eun-Kyung Kim, H. Moon, and J. Kwak. “Thyroid Nodules: Nondiagnostic Cytologic Results According to Thyroid Imaging Reporting and Data System Before and After Application of the Bethesda System.” *Radiology*, 2015.
- J. H. Yoon, Hye Sun Lee, Eun-Kyung Kim, H. Moon, V. Y. Park, and J. Kwak. “Follow-up Strategies for Thyroid Nodules with Benign Cytology on US-Guided Fine Needle Aspiration: Malignancy Rates of Management Guidelines Using US Before and After the Era of the Bethesda System.” *Thyroid*, 2019.
- J. Rabaglia, Wareef Kabbani, Lucy B Wallace, Shelby Holt, L. Watumull, J. Pruitt, W. Snyder, and F. Nwariaku. “Effect of the Bethesda System for Reporting Thyroid Cytopathology on Thyroidectomy Rates and Malignancy Risk in Cytologically Indeterminate Lesions.” *Surgery*, 2010.
- J. Sauter, Heidi D. Lehrke, Xiaotun Zhang, Osamah T Al Badri, R. Rodríguez-Gutiérrez, D. Delivanis, Naykky M Singh Ospina, et al. “Assessment of The Bethesda System for Reporting Thyroid Cytopathology.” *American Journal of Clinical Pathology*, 2019.
- Karima Rai, and Gurdeep Singh. “Diagnostic Accuracy of The Bethesda System for Reporting Thyroid Cytopathology: An Institution Experience — 2023 Guidelines Update.” *The Guthrie Journal*, 2025.
- Kimberly M. Ramonell, N. Otori, Jason B Liu, Kelly L. McCoy, A. Furlan, M. Tublin, S. Carty, and L. Yip. “Changes in Thyroid Nodule Cytology Rates After Institutional Implementation of the Thyroid Imaging Reporting and Data System.” *Surgery*, 2022.
- L. Layfield, and Z. Baloch. “Categorical Systems for Reporting of Cytology Specimens: Following the Footsteps of Bethesda-like Reporting Systems.” *Diagnostic Cytopathology*, 2020.
- Lamya A. Atweh, A. Al-Hadidi, Jasmeet Singh, Reham Alzahrani, Kelly Kersey, Adam Bobbey, Robert P Hoffman, J. Aldrink, and Summit Shah. “Quality Improvement Methodology to Improve Standardized Reporting of Pediatric Thyroid Ultrasounds Using TI-RADS.” *Journal of Pediatric Surgery*, 2023.
- Lawrence Q. Wong, and Z. Baloch. “Analysis of the Bethesda System for Reporting Thyroid Cytopathology and Similar Precursor Thyroid Cytopathology Reporting Schemes.” *Advances in Anatomic Pathology*, 2012.
- M. Olson, Thiraphon Boonyarunnate, A. A. Atlinboga, and Syed Z. Ali. “‘Suspicious for Papillary Thyroid Carcinoma’ Before and After The Bethesda System for Reporting Thyroid Cytopathology: Impact of Standardized Terminology.” *Acta Cytologica*, 2013.
- Maixin Wu. “A Correlation Study Between Thyroid Imaging Report and Data Systems and the Bethesda System for Reporting Thyroid Cytology with Surgical Follow-up - an Ultrasound-trained Cytopathologist's Experience.” *Diagnostic Cytopathology*, 2020.
- P. Upadhyaya, S. Dhakal, Purbesh Adhikari, B. Adhikari, Dibika Khadka, and S. Niraula. “Histopathological Review of Diagnostic Categories of the Bethesda System for Reporting Thyroid Cytopathology – An Institutional Experience of 5 Years.” *Journal of Cytology*, 2019.
- S. Lopes, Bijal Shah, and C. Eloy. “Thyroid Cytology: The Reality Before and After the Introduction of Ultrasound Classification Systems for Thyroid Nodules.” *Endocrinología Diabetes y Nutrición*, 2022.
- S. Ravindra, E. K. French, and L. Yip. “MON-LB84 Validation of TI-RADS (Thyroid Imaging, Reporting and Data System) Follow-Up Recommendations.” *Journal of the Endocrine Society*, 2020.
- Syed Z. Ali, and E. Cibas. “The Bethesda System for Reporting Thyroid Cytopathology II.” *Acta Cytologica*, 2016.
- T. Piticchio, Gilles Russ, M. Radzina, F. Frasca, C. Durante, and P. Trimoli. “Head-to-Head Comparison of American, European, and Asian TIRADSs in Thyroid Nodule Assessment: Systematic Review and Meta-Analysis.” *European Thyroid Journal*, 2024.
- Thiraphon Boonyarunnate, M. Olson, and Syed Z. Ali. “‘Suspicious for a Follicular Neoplasm’ Before and After the Bethesda System for Reporting Thyroid Cytopathology: Impact of Standardized Terminology.” *Acta Cytologica*, 2013.
- Vera A. Kachko, G. Semkina, N. Platonova, V. Vanushko, and A. Abrosimov. “Diagnosis of Thyroid Neoplasms: State

- of the Art on 2018.” *Endocrine Surgery*, 2018.
- W. Inabnet, F. Palazzo, J. Sosa, Joshua Kriger, S. Aspinall, M. Barczyński, G. Doherty, et al. “Correlating the Bethesda System for Reporting Thyroid Cytopathology with Histology and Extent of Surgery: A Review of 21,746 Patients from Four Endocrine Surgery Registries Across Two Continents.” *World Journal of Surgery*, 2020.
- W. Thedinger, Easwer Raman, and J. Dhingra. “Comparative Study of ACR TI-RADS and ATA 2015 for Ultrasound Risk Stratification of Thyroid Nodules.” *Otolaryngology Head & Neck Surgery*, 2021.
- X. Hu, J. Wu, P. Seal, S. Ghaznavi, C. Symonds, S. Kinnear, and R. Paschke. “Improvement in Thyroid Ultrasound Report Quality with Radiologists’ Adherence to 2015 ATA or 2017 TIRADS: A Population Study.” *European Thyroid Journal*, 2022.
- Xiaoxian Li, Chuan Peng, Ying Liu, Yixin Hu, Liang Yang, Yiwen Yu, Hongyan Zeng, et al. “Modified ACR TI-RADS and Modified AI TI-RADS for Thyroid Nodules: A Multicenter Retrospective Study.” *Thyroid*, 2023.
- Yun Zhu, Hongxun Wu, Bo-Kai Huang, Xin Shen, Gangming Cai, and Xiaobo Gu. “BRAFV600E Mutation Combined with American College of Radiology Thyroid Imaging Report and Data System Significantly Changes Surgical Resection Rate and Risk of Malignancy in Thyroid Cytopathology Practice.” *Gland Surgery*, 2020.